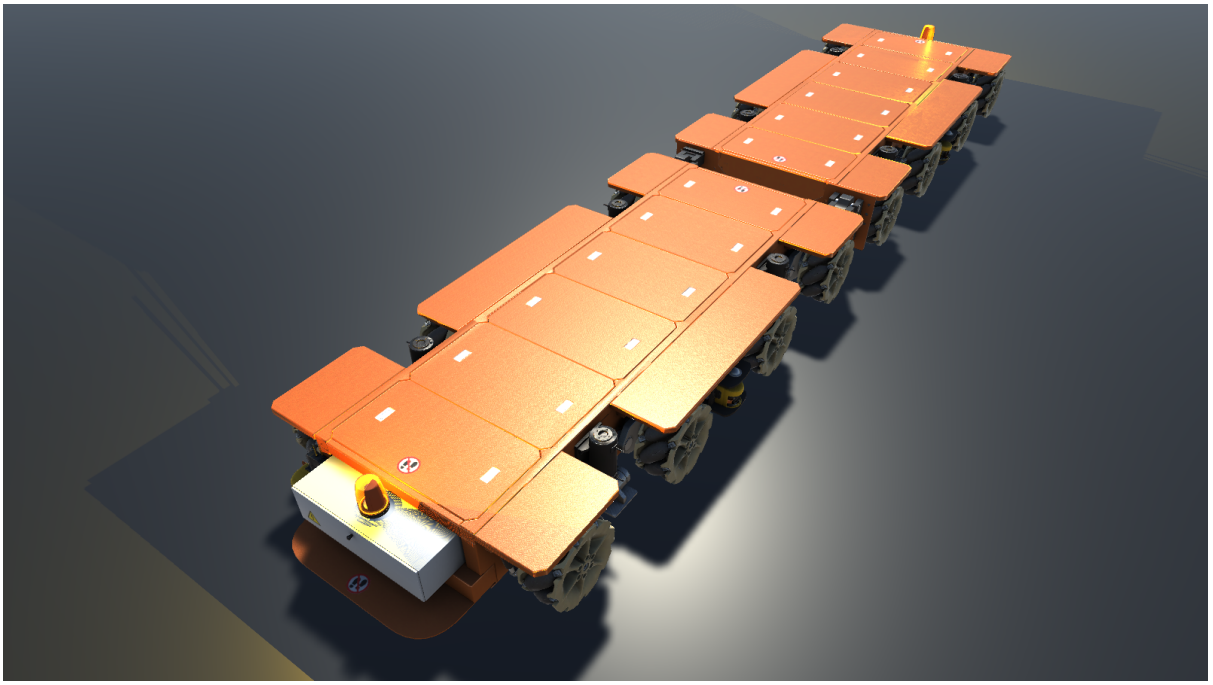
MEGAPOLY



AGV - OMNIWHEEL

Megapoly.art proud to introduce **"AGV - Omniwheel"**. This package contains an auto guided vehicle (AGV) with omniwheel(mecanum) capability. Omniwheel vehicles can have forward/lateral and diagonal movement as well as rotation on its axis. AGVs can be attached to other vehicles witch attach mechanism. This pack also includes separate beacon prefab.

The assets are optimized highpoly meshes with multi LODs combined with 4K textures.

Pack also includes controller scripts to help you understand the movement of the vehicle.

NOTE : Controller scripts included in demo scene are not optimized for performance and developed only for demo purposes !

Pack Info

Total : 10 Prefabs, 4 unique meshes

Release Log

1.0 : Initial release

How to Install

For the users using the **2019.4 Unity version or below**. Open the Asset Store inside Unity and download and import the Mechanic Shop Pack into your project.

For the users using the **2020.1 Unity version and above**. Open Asset Store in browser, press the "Add to my Assets" button and then the "Open in Unity" button. Go to the Unity Editor and import your asset via **Package Manager/My Assets**.

How to Use

After importing all the assets to your scene, you can access the prefabs under the Prefabs folder and directly start usşng in your scene.

There is one main material which can be used for all prefabs and this includes one texture. Alternatively you can check other materials inside the Materials folder which has prefix “offset”. This material is basically using the same texture with offset to get different color schemes.

Basic Controls (Demo Scene)

To control vehicle scripts use below keymapping :

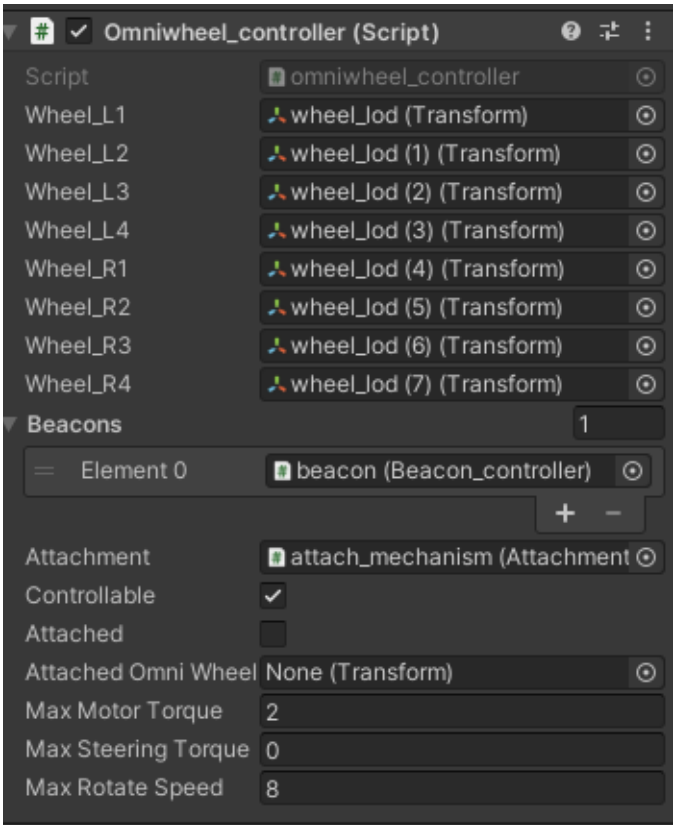
Up Arrow : Forward
Down Arrow : Backward
Left Arrow : Lateral left movement
Right Arrow : Lateral right movement
Q : Rotate clockwise
E : Rotate counter-clockwise
T : Detach
B : Beacons on/off

To attach another omniwheel, match attachment mechanisms and get close enough to attach other.

Scripts

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omniwheel_controller : This script is for controlling vehicle with keyboard. Requires rigidbody and collider on frame body. Basic movement of vehicle is done through rigidbody speed. Wheel rotations are based on mecanum dynamics.

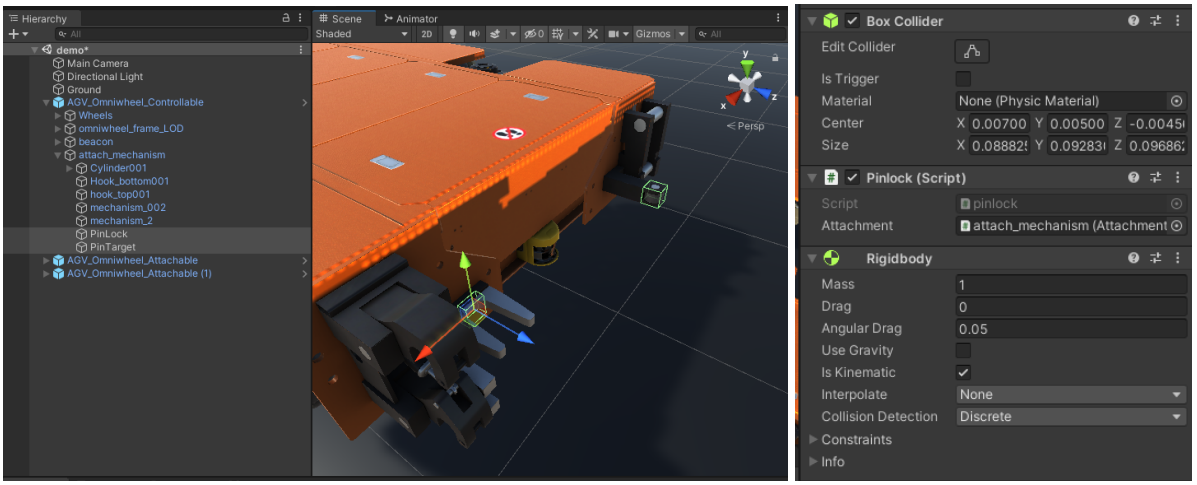


Wheels [Transform] : Attach 8 wheels in total 4 left, 4 right transforms
Beacons [List<beacon_controller>] : List of beacons on vehicle
Attachment [attachment_controller] : Attachment mechanism with controller script assigned.
Controllable [bool] : Check if you want to control the vehicle with keyboard, leave unchecked for other vehicles. Will be true when attached to main vehicle.
Attached [bool] : True when vehicle is attached, false when detached.
Attached Omni Wheel [Transform] : When another vehicle is attached, this slot will be assigned with other omni's transform.
Max Motor Torque [float] : Variable to control vehicle forward movement
Max Steering Torque [float] : Variable to control vehicle lateral movement
Max Rotate Speed [float] : Variable to control vehicle turn speed

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pinlock : This script is trigger check script to detect if there is close enough vehicle to attach. For this purpose this script triggers lock procedure.

The script should be attached to an empty gameobject with collider and should be positioned as below with corresponding trigger collider “pintarget”.



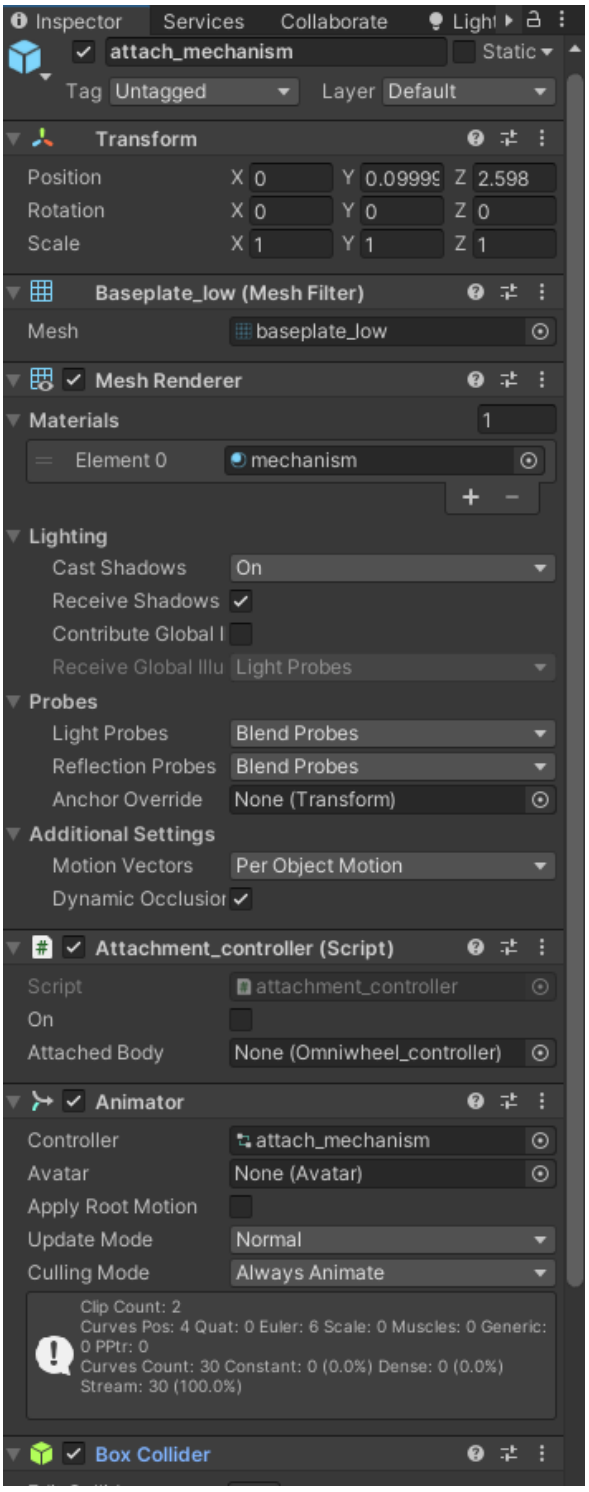
This setup should be child of attach mechanism.

Pinlock object should have collider and a rigidbody in order to run properly.

Attachment [attachment_controller] : Assign parent attachment mechanism

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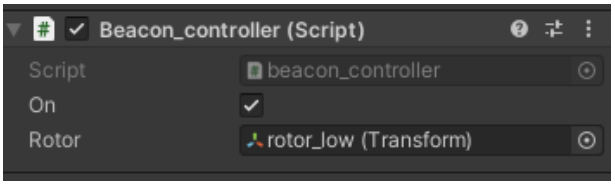
attachment_controller : This script is for controlling attached vehicle parameters. Attachment mechanism requires animator.



On [bool] : True/false value for attachment on or off.

Attached Body [omniwheel_controller] : Omni controller script parent of attachment mechanism.

beacon_controller : Beacon control script with on/off trigger.



On [bool] : True/false value for beacon on or off.
Rotor [Transform] : Rotating part of the beacon including point light.

Support

Check our other assets and services on our website.

www.megapoly.art

Please let us know if you have any questions.

info@megapoly.art

Enjoy,

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